

PROJECT JAMES

A local-first, auditable knowledge-reasoning platform — Graph-RAG retrieval, deterministic contradiction arbitration, and byte-identical replayable state.

v0.4.1 · T6 Causality Chain MIT License Python 3.11+ 3290 tests · all green DOI 10.5281/zenodo.2042671 Mother platform → v1.0

WHY DEVELOPERS LOVE JAMES

Local-first

Runs fully on-prem; no data leaves the box by default.

Replayable RAG

reconstruct_view_at() rebuilds any past state byte-identically.

Append-only audit

Every decision logged immutably — full provenance.

Deterministic arbiter

LLM-free 4-rule contradiction tree, replay-safe.

Plugin API

Typed module boundaries; extend without forking core.

Default-OFF invariant

New routing layers ship OFF — byte-identical until opt-in.

WHY JAMES ≠ FLAT / CLOUD RAG

- Replayable: past state reconstructed exactly, not approximated.
- Deterministic contradiction arbitration vs prompt-only heuristics.
- Explicit Trust Zones isolate low-trust input before the LLM.
- On-prem ownership + correction moat — no vendor lock-in.
- Cloud egress gated by an abstraction trust contract.

JAMES ARCHITECTURE OVERVIEW

ENTRY POINTS

Frontend / API

UI + REST surface

Auth + PolicyEngine

RBAC / ABAC · single policy source

Query Router

intent classify · scope routing

JAMES CORE

Hybrid Retrieval

vector + BM25 · rerank

Graph Engine

Graph-RAG · typed paths

Cognitive Middleware

auto-router · adaptive budget · entity-anchor · query-rewrite

Reasoning Loop

plan → retrieve → verify → synth

Trust-gated Memory

episodic · working · loom

EXECUTION SURFACES

Tool Router

FS / Web sandbox · allowlist

Cloud Egress Zone

abstraction module · mask + audit

Wiki Generator

auditable knowledge pages

Output Filter

PII + role mask

STATE & KNOWLEDGE

Layer 4 Lifecycle

T1 temporal · T7 supersede · T2 contradiction · T6 causality

Knowledge Cascade (L3)

delete / modify propagation across derived facts

Ontology + Typed Filter

relation schema · entity-typed graph filtering

Graph Snapshot / Replay

reconstruct_view_at — 'what was true at T?'

Append-only Audit Log

every state change immutable + replayable

HOW JAMES WORKS

1

Route

Query classified; scope + budget decided.

2

Retrieve

Hybrid vector+BM25 retrieval, reranked.

3

Traverse

Graph engine walks typed paths for evidence.

4

Reason

Loop plans, verifies, synthesizes ↔ memory.

5

Arbitrate

Deterministic 4-rule contradiction tree.

6

Mask

Output filtered for PII + role.

7

Audit

Append-only log records every decision.

PLATFORM CAPABILITIES (mother platform → v1.0)

- Auditable Graph-RAG over private corpora with full provenance.
- Time-travel queries — reconstruct knowledge as of any past point.
- Self-evolution with a mandatory human approval gate (opt-in only).
- Hybrid cloud reasoning tier behind an abstraction trust contract.
- v0.5 candidate: enterprise internal-knowledge ontology (gated).

WHY JAMES IS SPECIAL

- Replayable RAG — byte-identical historical state, by construction.
- Deterministic contradiction arbitration — no LLM in the decision path.
- Trust Zones gate every low-trust source before the LLM context.
- Human approval gate on self-evolution — never silent auto-deploy.
- Reproducible & cited — every headline number runs from a command.

WHAT'S VERIFIED (reproducible from current main)

3290

tests collected · all green on CI

1.00

QVT path_recall (median, paired)

0.58

graded_answer baseline

0.67

abstention F1

20 KB

hard module-size cap on core/

4-rule

LLM-free contradiction tree